Problem A. log2(N)

Time Limit 2000 ms
Mem Limit 1048576 kB

Problem Statement

Given a positive integer N, find the maximum integer k such that $2^k \leq N$.

Constraints

- N is an integer satisfying $1 \leq N \leq 10^{18}$.

Input

Input is given from Standard Input in the following format:

N

Output

Print the answer as an integer.

Sample 1

Input	Output
6	2

- $k = 2$ satisfies $2^2 = 4 \leq 6$.
- For every integer k such that $k \geq 3$, $2^k > 6$ holds.

Therefore, the answer is $k = 2$.

Sample 2

Input	Output
1	0

Note that $2^0 = 1$.

Sample 3

Input	Output
1000000000000000000	59

The input value may not fit into a 32-bit integer.

Problem B. Bitwise Exclusive Or

Time Limit 2000 ms

Problem Statement

You are given integers A and B between 0 and 255 (inclusive). Find a non-negative integer C such that $A \text{ xor } C = B$.

It can be proved that there uniquely exists such C, and it will be between 0 and 255 (inclusive).

▶ What is bitwise XOR?

Constraints

- $0 \leq A, B \leq 255$
- All values in input are integers.

Input

Input is given from Standard Input in the following format:

A B

Output

Print the answer.

Sample 1

Input	Output
3 6	5

When written in binary, 3 will be 11, and 5 will be 101. Thus, their xor will be 110 in binary, or 6 in decimal.

In short, $3 \text{ xor } 5 = 6$, so the answer is 5.

Sample 2

Input	Output
10 12	6

$$
\begin{array}{lrcr}
A & 1010 & \cdots & 10 \\
C & 110 & \cdots & 6 \\
\hline
B & 1100 & \cdots & 12
\end{array}
$$

Problem C. 1-2-4 Test

Time Limit 2000 ms
Mem Limit 1048576 kB

Problem Statement

There was an exam consisting of three problems worth 1, 2, and 4 points.

Takahashi, Aoki, and Snuke took this exam. Takahashi scored A points, and Aoki scored B points.

Snuke solved all of the problems solved by at least one of Takahashi and Aoki, and failed to solve any of the problems solved by neither of them.

Find Snuke's score.

It can be proved that Snuke's score is uniquely determined under the Constraints of this problem.

Constraints

- $0 \leq A, B \leq 7$
- A and B are integers.

Input

The input is given from Standard Input in the following format:

```
A B
```

Output

Print Snuke's score as an integer.

Sample 1

Input	Output
1 2	3

Since Takahashi scored 1 point, we see that he solved only the 1-point problem and failed to solve the other two.
Similarly, since Aoki scored 2 points, we see that he solved only the 2-point problem and failed to solve the other two.

Therefore, Snuke must have solved the 1- and 2-point problems, but not the 4-point one, which Takahashi and Aoki both failed to solve, for a score of 3 points. Thus, 3 should be printed.

Sample 2

Input	Output
5 3	7

Since Takahashi scored 5 points, we see that he solved the 1- and 4-point problems but not the 2-point one.
Similarly, since Aoki scored 3 points, we see that he solved the 1- and 2-point problems but not the 4-point one.

Therefore, each of the three problems is solved by at least one of Takahashi and Aoki, so we see that Snuke solved all of the problems, for a score of 7 points. Thus, 7 should be printed.

Sample 3

Input	Output
0 0	0

Both Takahashi and Aoki solved none of the problems. Therefore, so did Snuke. Thus, 0 should be printed.

Problem D. Submask

Time Limit 2000 ms
Mem Limit 1048576 kB

Problem Statement

You are given a non-negative integer N. Print all non-negative integers x that satisfy the following condition in ascending order.

- The set of the digit positions containing 1 in the binary representation of x is a subset of the set of the digit positions containing 1 in the binary representation of N .
  - That is, the following holds for every non-negative integer k: if the digit in the "2^ks" place of x is 1, the digit in the 2^ks place of N is 1.

Constraints

- N is an integer.
- $0 \leq N < 2^{60}$
- In the binary representation of N, at most 15 digit positions contain 1.

Input

The input is given from Standard Input in the following format:

```
N
```

Output

Print the answer as decimal integers in ascending order, each in its own line.

Sample 1

Input	Output
11	0 1 2 3 8 9 10 11

The binary representation of $N = 11_{(10)}$ is $1011_{(2)}$.
The non-negative integers x that satisfy the condition are:

- $0000_{(2)} = 0_{(10)}$
- $0001_{(2)} = 1_{(10)}$
- $0010_{(2)} = 2_{(10)}$
- $0011_{(2)} = 3_{(10)}$
- $1000_{(2)} = 8_{(10)}$
- $1001_{(2)} = 9_{(10)}$
- $1010_{(2)} = 10_{(10)}$
- $1011_{(2)} = 11_{(10)}$

Sample 2

Input	Output
0	0

Sample 3

Input	Output
576461302059761664	0 524288 549755813888 549756338176 576460752303423488 576460752303947776 576461302059237376 576461302059761664

The input may not fit into a 32-bit signed integer.

Problem E. Play with Binary Numbers

Time Limit	1201 ms
Mem Limit	1572864 kB
Code Length Limit	50000 B
OS	Linux

Let S be the binary representation of an integer. We define two functions a(i) and b(i) such that:

- a(i) = Number of occurrences of '1' at odd positions of S,
- b(i) = Number of occurrences of '1' at even positions of S.

For example: for integer 19, S = 10011, so a(19) = 2 and b(19) = 1.

Input

First line contains an integer T, the number of test cases. Then T lines follow. On each line, you will be given three integers M, N, K.

Output

For each test case output a single integer R, where R is the number of integers 'i' between M and N (both inclusive) such that absolute difference of a(i) and b(i) is equal to K. The answer to each each test case should be on a separate line.

Constraints

$T \le 50$

$1 \le M < N \le 10^{19}$

$1 \le N - M \le 10^{6}$

$0 \le K \le 50$

Example

Input	Output
1 1 10 2	2

Problem F. Bitmask

Time Limit 2000 ms

Problem Statement

You are given an integer N and a string S consisting of `0`, `1`, and `?`. Let T be the set of values that can be obtained by replacing each `?` in S with `0` or `1` and interpreting the result as a binary integer. For instance, if $S =$ `?0?`, we have $T = \{000_{(2)}, 001_{(2)}, 100_{(2)}, 101_{(2)}\} = \{0, 1, 4, 5\}$.

Print (as a decimal integer) the greatest value in T less than or equal to N. If T does not contain a value less than or equal to N, print `-1` instead.

Constraints

- S is a string consisting of `0`, `1`, and `?`.
- The length of S is between 1 and 60, inclusive.
- $1 \leq N \leq 10^{18}$
- N is an integer.

Input

The input is given from Standard Input in the following format:

S
N

Output

Print the answer.

Sample 1

Input	Output
?0? 2	1

As shown in the problem statement, $T = \{0, 1, 4, 5\}$. Among them, 0 and 1 are less than or equal to N, so you should print the greatest of them, 1.

Sample 2

Input	Output
101 4	-1

We have $T = \{5\}$, which does not contain a value less than or equal to N.

Sample 3

Input	Output
?0? 1000000000000000000	5

Problem G. ORXOR

Time Limit 2000 ms
Mem Limit 1048576 kB

Problem Statement

Given is a number sequence A of length N.
Let us divide this sequence into one or more non-empty contiguous intervals.
Then, for each of these intervals, let us compute the bitwise OR of the numbers in it.
Find the minimum possible value of the bitwise XOR of the values obtained in this way.

▶ What is bitwise OR?

▶ What is bitwise XOR?

Constraints

- $1 \leq N \leq 20$
- $0 \leq A_i < 2^{30}$
- All values in input are integers.

Input

Input is given from Standard Input in the following format:

N
$A_1 \ A_2 \ A_3 \ \ldots \ A_N$

Output

Print the answer.

Sample 1

Input	Output
3 1 5 7	2

If we divide $[1, 5, 7]$ into $[1, 5]$ and $[7]$, their bitwise ORs are 5 and 7, whose XOR is 2. It is impossible to get a smaller result, so we print 2.

Sample 2

Input	Output
3 10 10 10	0

We should divide this sequence into $[10]$ and $[10, 10]$.

Sample 3

Input	Output
4 1 3 3 1	0

We should divide this sequence into $[1, 3]$ and $[3, 1]$.

Problem H. XOR World

Time Limit 2000 ms
Mem Limit 1048576 kB

Problem Statement

Let $f(A, B)$ be the exclusive OR of $A, A+1, \ldots, B$. Find $f(A, B)$.

▶ What is exclusive OR?

Constraints

- All values in input are integers.
- $0 \leq A \leq B \leq 10^{12}$

Input

Input is given from Standard Input in the following format:

A B

Output

Compute $f(A, B)$ and print it.

Sample 1

Input	Output
2 4	5

$2, 3, 4$ are `010`, `011`, `100` in base two, respectively. The exclusive OR of these is `101`, which is 5 in base ten.

Sample 2

Input	Output
123 456	435

Sample 3

Input	Output
123456789012 123456789012	123456789012

Problem I. AND and SUM

Time Limit 2000 ms
Mem Limit 1048576 kB

Problem Statement

Solve the following problem for T test cases.

> Given are non-negative integers a and s. Is there a pair of non-negative integers (x, y) that satisfies both of the conditions below?
>
> - $x \text{ AND } y = a$
> - $x + y = s$

▶ What is bitwise AND?

Constraints

- $1 \leq T \leq 10^5$
- $0 \leq a, s < 2^{60}$
- All values in input are integers.

Input

Input is given from Standard Input. The first line is in the following format:

```
T
```

Then, T test cases follow. Each test case is in the following format:

```
a s
```

Output

Print T lines. The i-th line $(1 \leq i \leq T)$ should contain `Yes` if, in the i-th test case, there is a pair of non-negative integers (x, y) that satisfies both of the conditions in the

Problem Statement, and `No` otherwise.

Sample 1

Input	Output
2 1 8 4 2	Yes No

In the first test case, some pairs such as $(x, y) = (3, 5)$ satisfy the conditions.

In the second test case, no pair of non-negative integers satisfies the conditions.

Sample 2

Input	Output
4 201408139683277485 381410962404666524 360288799186493714 788806911317182736 18999951915747344 451273909320288229 962424162689761932 1097438793187620758	No Yes Yes No

Carl is right now the happiest child in the world: he has just learned this morning what the binary system is. He learned, for instance, that the binary representation of a positive integer k is a string $a_n a_{n-1} \ldots a_1 a_0$ where each a_i is a binary digit 0 or 1, starting with $a_n = 1$, and such that $k = \Sigma_{i=0}^{n} a_i \times 2^i$. It is really nice to see him turning decimal numbers into binary numbers, and then adding and even multiplying them.

Caesar is Carl's older brother, and he just can't stand to see his little brother so happy. So he has prepared a challenge: "Look Carl, I have an easy question for you: I will give you two integers A and B, and you have to tell me how many 1's there are in the binary representation of all the integers from A to B, inclusive. Get ready". Carl agreed to the challenge. After a few minutes, he came back with a list of the binary representation of all the integers from 1 to 100. "Caesar, I'm ready". Caesar smiled and said: "Well, let me see, I choose $A = 10^{15}$ and $B = 10^{16}$. Your list will not be useful".

Carl hates loosing to his brother so he needs a better solution fast. Can you help him?

Input

The input file contains several test cases, each of them as described below.

A single line that contains two integers A and B ($1 \leq A \leq B \leq 10^{16}$).

Output

For each test case, output a line with an integer representing the total number of digits 1 in the binary representation of all the integers from A to B, inclusive.

Sample Input

```
1000000000000000 10000000000000000
2 12
9007199254740992 9007199254740992
```

Sample Output

```
239502115812196372
21
1
```